

Deep Learning and Its Application

Week 11: Practicals

The goals of this practice are to have a clear understanding of some key concepts about reinforcement learning and implementation of Grid World-Q. You need to research by reading web materials before you play the program and answer the questions. You need to write your answers to E1.2 in a MS-Word file for your own exercises.

E1.1 Labwork:

- 1) Read Reinforcement Learning – Implement Grid World-Q at <https://medium.com/data-science/implement-grid-world-with-q-learning-51151747b455>
- 2) Download the code and create a Python file with the name of “gridWorld_Q” using PyCharm, cut and past the code into the gridWorld_Q
- 3) Run the gridWorld_Q
- 4) Understand next state is chosen
- 5) Understand how an action is non-deterministic
- 6) Look at the slides 12-20 of lecture Week 4, how Q-values are calculated and updated
- 7) Compare the previous version of gridWorld with gridWorld_Q on the data structure and calculation of Q-values
- 8) Change hyper parameters in the code and understand their impact on the game

E1.2 based on the game to define in your own words: (a) agent (b) environment (c) state (d) action (e) pairs of state-action (f) Q-values